

# A Rigorous Continuous-Time Characterization of the IID Multi-Selection Prophet Ratio

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## Abstract

For fixed inventory  $k$ , let  $\gamma_{n,k}$  be the worst-case ratio of the optimal online policy to the prophet for  $n$  independent and identically distributed nonnegative observations, when at most  $k$  observations may be selected. We prove that  $\gamma_{n,k}$  has a limit as  $n \rightarrow \infty$  and identify that limit as the value of an explicit continuous-time max–min control problem. We then prove an exact minimax duality formula. The dual is a probability measure on the Poisson scale, and its support function is computed by a finite-state Hamilton–Jacobi–Bellman equation. A compactification at scales zero and infinity yields an attained measure dual; the mass at infinity can be removed. Rewriting the positive-scale measure by its weighted tail gives the exact variational formula

$$\rho_k = \min_{(\kappa, H) \in \mathfrak{H}_k} u_k^{\kappa, H}(0), \quad -\dot{u}_\ell = \sup_{a \geq 0} \left\{ \kappa + \int_0^a H(r) \, dr - a(u_\ell - u_{\ell-1}) \right\},$$

where the displayed equation is understood as the increasing limit of bounded-action value equations. We also show that every fixed-horizon value  $\gamma_{n,k}$  is a real algebraic number.

The result is *substantial partial progress*, not a complete solution of the open questions in Brustle–Perez-Salazar–Verdugo. In particular, we do not prove that their nonlinear splitting system is tight, that  $\inf_{n \geq k} \gamma_{n,k} = \lim_{n \rightarrow \infty} \gamma_{n,k}$ , or that the regular free-boundary system derived here has a smooth full-support optimizer for  $k \geq 2$ .

**Status.** The mathematical gaps in the earlier draft can be repaired far enough to prove an exact asymptotic variational and Hamiltonian characterization. This is a genuine result about the problem studied in [2], but it does not settle the authors’ principal tightness question. The correct classification is therefore **partial progress, not a complete solution**.

## 1 Problem, notation, and contribution

Let  $X_1, \dots, X_n$  be IID, nonnegative, and drawn from a distribution  $F$  known to the decision maker. A sequential policy may accept at most  $k$  observations. Write

$$\text{DP}_{n,k}(F) := \sup_{\pi} \mathbb{E}_F \left[ \sum_{i \in \pi} X_i \right], \quad \text{OPT}_{n,k}(F) := \mathbb{E}_F [\text{sum of the largest } k \text{ among } X_1, \dots, X_n],$$

and define

$$\gamma_{n,k} := \inf_F \frac{\text{DP}_{n,k}(F)}{\text{OPT}_{n,k}(F)}. \tag{1}$$

The exact type-coverage formulation of Jiang–Ma–Zhang [4] is our finite-horizon starting point.

For  $q \in [0, 1]$  and  $s \geq 0$ , set

$$B_{n,k}(q) := \mathbb{E}[\min\{\text{Bin}(n, q), k\}], \quad (2)$$

$$Q_k(s) := \mathbb{E}[\min\{\text{Pois}(s), k\}], \quad (3)$$

$$w_k(s) := Q'_k(s) = \mathbb{P}\{\text{Pois}(s) \leq k-1\} = e^{-s} \sum_{j=0}^{k-1} \frac{s^j}{j!}. \quad (4)$$

Thus

$$Q_k(s) = \int_0^s w_k(r) dr, \quad \frac{Q_k(s)}{s} \longrightarrow 1 \quad (s \downarrow 0), \quad \frac{Q_k(s)}{k} \longrightarrow 1 \quad (s \rightarrow \infty). \quad (5)$$

Two corrections are decisive.

First, pointwise closure is too weak for a max–min objective: a bad scale can move with the approximating sequence. We instead use uniform convergence on compact scale intervals and prove quantitative inequalities controlling all smaller and larger scales.

Second, compactness alone does not imply existence of a saddle when the payoff lacks the required semicontinuity. We first prove minimax and attainment on every compact scale interval, and only then take weak limits of genuine compact-scale minimizers.

A third, subtler issue is convexification. The continuous feasible class contains ex ante mixtures of controls. A mixture cannot simply be replaced by one of its components. Lemma 3.2 constructs a single bounded Markov hazard whose coverage dominates the entire mixture pointwise.

## 2 The exact finite-horizon formulation

For  $i \in [n]$  and  $\ell \in [k]$ , let  $x_i^\ell$  be the probability that exactly  $\ell$  slots remain immediately before observation  $i$ , and let  $y_i^\ell$  be the unconditional probability of accepting observation  $i$  in that state. Put  $y_i^{k+1} = 0$ . Let  $\mathcal{X}_{n,k}$  be the compact polytope determined by

$$x_1^k = 1, \quad x_1^\ell = 0 \quad (\ell < k), \quad (6)$$

$$x_{i+1}^\ell = x_i^\ell - y_i^\ell + y_i^{\ell+1} \quad (i \in [n-1], \ell \in [k]), \quad (7)$$

$$0 \leq y_i^\ell \leq x_i^\ell. \quad (8)$$

For convenience, define  $x_{n+1}$  by the same flow equation at  $i = n$ .

**Theorem 2.1** (Jiang–Ma–Zhang type coverage). *For  $n > k$ ,*

$$\gamma_{n,k} = \max_{(x,y) \in \mathcal{X}_{n,k}} \inf_{q \in (0,1]} \frac{\sum_{i=1}^n \sum_{\ell=1}^k \min\{y_i^\ell, qx_i^\ell\}}{B_{n,k}(q)}. \quad (9)$$

*For  $n = k$ , one has  $\gamma_{n,k} = 1$ .*

This is the IID dynamic-policy/prophet semi-infinite program of [4]. We record two exact consequences.

## 2.1 A finite measure–Bellman dual

For a probability measure  $\lambda$  on  $(0, 1]$ , define

$$\Phi_{n,k}^\lambda(p) := \int_{(0,1]} \frac{\min\{p, q\}}{B_{n,k}(q)} \lambda(\mathrm{d}q), \quad 0 \leq p \leq 1. \quad (10)$$

Let  $V_{n+1,\ell}^\lambda = 0$ ,  $V_{i,0}^\lambda = 0$ , and recursively set

$$V_{i,\ell}^\lambda = V_{i+1,\ell}^\lambda + \max_{0 \leq p \leq 1} \left\{ \Phi_{n,k}^\lambda(p) - p(V_{i+1,\ell}^\lambda - V_{i+1,\ell-1}^\lambda) \right\}. \quad (11)$$

**Theorem 2.2** (Exact finite Bellman dual). *For every  $n > k$ ,*

$$\gamma_{n,k} = \inf_{\lambda \in \mathcal{P}((0,1])} V_{1,k}^\lambda = \inf_{\lambda \in \mathcal{P}_{\text{fin}}((0,1])} V_{1,k}^\lambda, \quad (12)$$

where  $\mathcal{P}_{\text{fin}}$  denotes finitely supported probability measures. No attainment assertion is made.

*Proof.* For  $(x, y) \in \mathcal{X}_{n,k}$ , write

$$f_{x,y}(q) := \frac{\sum_{i,\ell} \min\{y_i^\ell, qx_i^\ell\}}{B_{n,k}(q)}.$$

For every  $q > 0$ , the map  $(x, y) \mapsto f_{x,y}(q)$  is continuous and concave. It is uniformly bounded: the numerator is at most  $nq$ , while  $B_{n,k}(q) \geq \mathbb{P}\{\text{Bin}(n, q) \geq 1\} = 1 - (1 - q)^n \geq q$ .

Equip probability measures with total variation. For each fixed  $(x, y)$ , integration of the bounded function  $f_{x,y}$  is continuous and affine in the measure. Sion's theorem [7], first with  $\mathcal{P}((0, 1])$  and then with the convex set  $\mathcal{P}_{\text{fin}}((0, 1])$ , gives

$$\max_{(x,y) \in \mathcal{X}_{n,k}} \inf_{\lambda} \int f_{x,y} \mathrm{d}\lambda = \inf_{\lambda} \max_{(x,y) \in \mathcal{X}_{n,k}} \int f_{x,y} \mathrm{d}\lambda.$$

On the left, the infimum over either class of measures is  $\inf_q f_{x,y}(q)$  because all Dirac measures are available. Hence both right-hand infima equal  $\gamma_{n,k}$ .

Fix  $\lambda$ . When  $x_i^\ell > 0$ , put  $p_i^\ell = y_i^\ell / x_i^\ell$ ; define it arbitrarily when  $x_i^\ell = 0$ . Then

$$\int f_{x,y}(q) \lambda(\mathrm{d}q) = \sum_{i=1}^n \sum_{\ell=1}^k x_i^\ell \Phi_{n,k}^\lambda(p_i^\ell).$$

This is the total reward of a finite-state Markov decision process in which action  $p$  moves from state  $\ell$  to  $\ell - 1$  with probability  $p$  and earns current reward  $\Phi_{n,k}^\lambda(p)$ . Backward induction is exactly (11); its initial value is  $V_{1,k}^\lambda$ .  $\square$

## 2.2 Algebraicity

**Theorem 2.3** (Every fixed-horizon ratio is algebraic). *For every fixed  $n$  and  $k$ , the number  $\gamma_{n,k}$  is real algebraic. More precisely, it is the largest  $\theta$  satisfying a first-order formula over the real closed field with rational coefficients; quantifier elimination therefore gives a terminating exact symbolic procedure.*

*Proof.* For fixed  $n, k$ , the polynomial

$$B_{n,k}(q) = \sum_{j=0}^n \min\{j, k\} \binom{n}{j} q^j (1-q)^{n-j}$$

has rational coefficients. By Theorem 2.1, a number  $\theta$  is feasible if and only if there exist  $(x, y) \in \mathcal{X}_{n,k}$  such that, for every  $q \in (0, 1]$ , there exist auxiliary variables  $z_i^\ell$  satisfying

$$0 \leq z_i^\ell \leq y_i^\ell, \quad z_i^\ell \leq qx_i^\ell, \quad \theta B_{n,k}(q) \leq \sum_{i,\ell} z_i^\ell.$$

For fixed  $q$ , such  $z$  exist exactly when  $\theta B_{n,k}(q) \leq \sum_{i,\ell} \min\{y_i^\ell, qx_i^\ell\}$ . Thus feasibility is a first-order formula involving only polynomial equalities and inequalities with rational coefficients. Tarski–Seidenberg elimination [1] implies that the feasible set of  $\theta$  is semialgebraic over  $\mathbb{Q}$ .

The objective in (9) attains its maximum: the pointwise infimum of the continuous functions  $(x, y) \mapsto f_{x,y}(q)$  is upper semicontinuous on the compact set  $\mathcal{X}_{n,k}$ . A bounded one-dimensional semialgebraic set over  $\mathbb{Q}$  is a finite union of points and intervals whose endpoints are real algebraic. Its largest point,  $\gamma_{n,k}$ , is therefore algebraic. The case  $n = k$  is immediate.  $\square$

**Remark 2.4.** *Theorem 2.3 is an exact finite description, but not a tractable analytic formula of the kind sought in [2]; generic quantifier elimination has enormous complexity.*

### 3 Continuous-time coverage profiles

A bounded hazard control is a  $k$ -tuple  $a = (a_1, \dots, a_k)$  of bounded nonnegative measurable functions on  $[0, 1]$ . Its state probabilities solve

$$\dot{x}_\ell(t) = -a_\ell(t)x_\ell(t) + a_{\ell+1}(t)x_{\ell+1}(t), \quad \ell = 1, \dots, k, \quad (13)$$

$$x_k(0) = 1, \quad x_\ell(0) = 0 \quad (\ell < k), \quad a_{k+1}x_{k+1} \equiv 0. \quad (14)$$

Its Poisson-scale coverage is

$$C_a(s) := \sum_{\ell=1}^k \int_0^1 x_\ell(t) \min\{a_\ell(t), s\} dt, \quad s \geq 0. \quad (15)$$

Writing  $m_\ell = x_\ell a_\ell$ , the integrand is  $\min\{m_\ell, sx_\ell\}$ .

**Lemma 3.1** (Shape of a coverage profile). *For every bounded hazard control,  $C_a$  is nonnegative, nondecreasing, concave, satisfies  $C_a(0) = 0$ , is 1-Lipschitz, and is bounded by  $k$ .*

*Proof.* All assertions except the Lipschitz bound follow directly from (15). For  $0 \leq r < s$ ,

$$0 \leq C_a(s) - C_a(r) \leq (s - r) \int_0^1 \sum_{\ell=1}^k x_\ell(t) dt \leq s - r,$$

because at every time the process occupies at most one positive-inventory state. Finally,  $C_a(s) \leq \sum_{\ell} \int x_\ell a_\ell \leq k$ , since the latter is the expected number of transitions.  $\square$

**Lemma 3.2** (Markovization dominates ex ante mixing). *Let  $a^1, \dots, a^m$  be bounded hazard controls and let  $\alpha_r \geq 0$ ,  $\sum_r \alpha_r = 1$ . There is a bounded hazard control  $a$  such that*

$$C_a(s) \geq \sum_{r=1}^m \alpha_r C_{a^r}(s) \quad \text{for every } s \geq 0. \quad (16)$$

*Proof.* Let  $x^r$  be the state under  $a^r$  and  $m_\ell^r = x_\ell^r a_\ell^r$ . Put

$$x_\ell = \sum_r \alpha_r x_\ell^r, \quad m_\ell = \sum_r \alpha_r m_\ell^r, \quad a_\ell = \begin{cases} m_\ell / x_\ell, & x_\ell > 0, \\ 0, & x_\ell = 0. \end{cases}$$

The pair  $(x, m)$  satisfies the same linear flow equations and initial conditions, so  $x$  is the state generated by  $a$ . If all original controls are bounded by  $A$ , then  $m_\ell \leq A x_\ell$ , hence  $a_\ell \leq A$ .

For fixed  $s$ , the function

$$(x, m) \mapsto \min\{m, sx\}$$

is concave on  $\mathbb{R}_+^2$ , being the minimum of two affine functions. Therefore

$$\min\{m_\ell, s x_\ell\} \geq \sum_r \alpha_r \min\{m_\ell^r, s x_\ell^r\}.$$

Integrating and summing over  $\ell$  proves (16).  $\square$

**Lemma 3.3** (Step controls are dense on compact scale sets). *For every bounded measurable control  $a$ , every  $S < \infty$ , and every  $\eta > 0$ , there is a bounded step control  $b$  such that*

$$\sup_{0 \leq s \leq S} |C_b(s) - C_a(s)| < \eta. \quad (17)$$

*The same conclusion holds when  $b_\ell = a_\ell + c$  with  $c \downarrow 0$ ; in particular, adding a small common positive hazard changes coverage locally uniformly by  $o(1)$ .*

*Proof.* Choose bounded step functions  $b_\ell$  converging to  $a_\ell$  in  $L^1[0, 1]$ , with a common uniform bound. The coefficient matrices in (13) then converge in  $L^1$ . Gronwall's inequality gives uniform convergence of the corresponding states. Since  $z \mapsto \min\{z, s\}$  is 1-Lipschitz and bounded by  $S$  for  $s \leq S$ , the difference of the coverage integrals is bounded by a constant times the  $L^1$  difference of the controls plus the uniform difference of the states, uniformly in  $s \in [0, S]$ . The perturbation  $a \mapsto a + c$  is the same argument.  $\square$

Let  $\mathcal{C}_k$  be the closure, under uniform convergence on compact subsets of  $[0, \infty)$ , of the convex hull of all bounded-control coverage functions. Lemma 3.1 and a diagonal Arzela–Ascoli argument imply that  $\mathcal{C}_k$  is compact and convex in the compact-open topology.

Lemma 3.2 has the following useful consequence: for every  $C \in \mathcal{C}_k$ , every compact interval  $K \subset [0, \infty)$ , and every  $\eta > 0$ , there is a single bounded hazard control  $a$  such that

$$C_a(s) \geq C(s) - \eta \quad (s \in K). \quad (18)$$

Indeed, approximate  $C$  on  $K$  by a finite convex combination and then apply the lemma.

**Lemma 3.4** (The one-scale envelope). *For every bounded control and every  $s \geq 0$ ,*

$$C_a(s) \leq Q_k(s). \quad (19)$$

*Consequently, every  $C \in \mathcal{C}_k$  satisfies  $C(s) \leq Q_k(s)$ .*

*Proof.* Fix  $s > 0$ . Let  $V_0(\tau) = 0$  and

$$V_\ell(\tau) := \mathbb{E}[\min\{\text{Pois}(s\tau), \ell\}], \quad \ell = 1, \dots, k.$$

Then

$$V_\ell(\tau) - V_{\ell-1}(\tau) = \mathbb{P}\{\text{Pois}(s\tau) \geq \ell\},$$

and a direct differentiation gives

$$\partial_\tau V_\ell(\tau) = s\mathbb{P}\{\text{Pois}(s\tau) \leq \ell - 1\}.$$

For  $d = V_\ell - V_{\ell-1} \in [0, 1]$ ,

$$\sup_{a \geq 0} \{\min\{a, s\} - ad\} = s(1 - d),$$

with an optimizer  $a = s$ . Thus  $V$  solves the dynamic programming equations for the one-scale reward in (15). Starting with  $k$  slots and horizon one, the optimal value is  $V_k(1) = Q_k(s)$ , proving (19). The case  $s = 0$  and passage to  $\mathcal{C}_k$  follow by continuity.  $\square$

**Definition 3.5** (The continuous Poisson-scale game). *Define*

$$\rho_k := \max_{C \in \mathcal{C}_k} \inf_{s > 0} \frac{C(s)}{Q_k(s)}. \quad (20)$$

The maximum is attained because  $C \mapsto \inf_{s > 0} C(s)/Q_k(s)$  is an infimum of continuous evaluation functionals and hence is upper semicontinuous. Lemma 3.4 gives  $0 \leq \rho_k \leq 1$ .

## 4 Uniform scale localization and finite embeddings

### 4.1 Two tail inequalities

**Lemma 4.1** (Uniform localization of the adversarial scale). *Let  $C : [0, \infty) \rightarrow [0, k]$  be nondecreasing and concave with  $C(0) = 0$ . Then, for every  $0 < s \leq \varepsilon$ ,*

$$\frac{C(s)}{Q_k(s)} \geq \frac{C(\varepsilon)}{\varepsilon}, \quad (21)$$

*and, for every  $s \geq M$ ,*

$$\frac{C(s)}{Q_k(s)} \geq \frac{C(M)}{k}. \quad (22)$$

*The same statements hold for a finite coverage profile after replacing  $Q_k(s)$  by  $B_{n,k}(s/n)$ , for  $0 < s \leq n$ .*

*Proof.* Concavity and  $C(0) = 0$  imply that  $C(s)/s$  is nonincreasing. Since  $Q_k(s) \leq s$ , for  $s \leq \varepsilon$ ,

$$\frac{C(s)}{Q_k(s)} \geq \frac{C(s)}{s} \geq \frac{C(\varepsilon)}{\varepsilon}.$$

For  $s \geq M$ , monotonicity of  $C$  and  $Q_k(s) \leq k$  give  $C(s)/Q_k(s) \geq C(M)/k$ . For the finite denominator use  $B_{n,k}(s/n) \leq \min\{s, k\}$ .  $\square$

## 4.2 Embedding finite policies

For  $(x, y) \in \mathcal{X}_{n,k}$ , define the scaled coverage

$$C_{n;x,y}(s) := \sum_{i=1}^n \sum_{\ell=1}^k \min \left\{ y_i^\ell, \frac{s}{n} x_i^\ell \right\}, \quad s \geq 0. \quad (23)$$

For  $0 < s \leq n$ , the objective in (9) is  $C_{n;x,y}(s)/B_{n,k}(s/n)$  under  $s = nq$ .

**Lemma 4.2** (Finite policies have continuous-time embeddings). *For every  $(x, y) \in \mathcal{X}_{n,k}$  there is a function  $\tilde{C}_{n;x,y} \in \mathcal{C}_k$  such that, for every  $S < \infty$ ,*

$$\sup_{0 \leq s \leq S} \left| \tilde{C}_{n;x,y}(s) - C_{n;x,y}(s) \right| \leq \frac{kS}{n}. \quad (24)$$

*Proof.* Put  $I_i = [(i-1)/n, i/n]$ . On  $I_i$ , define

$$m_\ell(t) := ny_i^\ell, \quad \tilde{x}_\ell(t) := x_i^\ell + n \left( t - \frac{i-1}{n} \right) (x_{i+1}^\ell - x_i^\ell).$$

The discrete flow equation gives  $\dot{\tilde{x}}_\ell = -m_\ell + m_{\ell+1}$ , and the interpolation is nonnegative. Define the generalized-flow coverage

$$\tilde{C}_{n;x,y}(s) := \sum_{\ell=1}^k \int_0^1 \min\{m_\ell(t), s\tilde{x}_\ell(t)\} dt. \quad (25)$$

We first prove that this function lies in  $\mathcal{C}_k$ .

Let  $z$  be the state process under the reference control with hazard 1 in every positive state, and let  $r_\ell = z_\ell$ . For  $0 < \delta < 1$ , set

$$x_\ell^\delta = (1 - \delta)\tilde{x}_\ell + \delta z_\ell, \quad m_\ell^\delta = (1 - \delta)m_\ell + \delta r_\ell, \quad a_\ell^\delta = \frac{m_\ell^\delta}{x_\ell^\delta}$$

when  $x_\ell^\delta > 0$ , and set  $a_\ell^\delta = 0$  otherwise. The pair  $(x^\delta, m^\delta)$  satisfies the exact flow equations and initial condition, hence  $x^\delta$  is generated by  $a^\delta$ .

The control  $a^\delta$  is bounded. For  $\ell = k$ ,  $z_k(t) = e^{-t}$  is bounded away from zero. For  $\ell < k$ , a discrete policy cannot have  $y_i^\ell > 0$  before at least  $k - \ell$  earlier selections are possible; hence  $m_\ell$  vanishes on a neighborhood of zero. On the remaining compact part of the interval, the reference state  $z_\ell(t) = e^{-t} t^{k-\ell} / (k - \ell)!$  is bounded away from zero wherever  $m_\ell$  can be positive. Where  $m_\ell = 0$ , the ratio is at most 1. Thus  $a^\delta$  is bounded for each fixed  $\delta$ .

For  $s \leq S$ , the elementary inequality

$$|\min\{m, sx\} - \min\{m', sx'\}| \leq |m - m'| + s|x - x'|$$

shows that the coverage of  $a^\delta$  converges to (25) uniformly on  $[0, S]$  as  $\delta \downarrow 0$ . Therefore  $\tilde{C}_{n;x,y} \in \mathcal{C}_k$ .

It remains to compare it with (23). Since  $z \mapsto \min\{ny_i^\ell, sz\}$  is  $s$ -Lipschitz,

$$\begin{aligned} \left| \tilde{C}_{n;x,y}(s) - C_{n;x,y}(s) \right| &\leq s \sum_{i,\ell} \int_{I_i} |\tilde{x}_\ell(t) - x_i^\ell| dt \\ &= \frac{s}{2n} \sum_{i,\ell} |x_{i+1}^\ell - x_i^\ell| \\ &\leq \frac{s}{2n} \sum_{i,\ell} (y_i^\ell + y_i^{\ell+1}) \leq \frac{ks}{n}. \end{aligned}$$

The last inequality uses  $\sum_{i,\ell} y_i^\ell \leq k$ , the expected number of selections. Taking  $s \leq S$  proves (24).  $\square$

**Lemma 4.3** (Uniform Poisson approximation). *For every fixed  $M < \infty$ ,*

$$\sup_{0 \leq s \leq M} |B_{n,k}(s/n) - Q_k(s)| \longrightarrow 0. \quad (26)$$

*Proof.* Le Cam's inequality [6] bounds the total variation distance between  $\text{Bin}(n, s/n)$  and  $\text{Pois}(s)$  by a constant times  $s^2/n$ . Since  $m \mapsto \min\{m, k\}$  is bounded by  $k$ , the expectation difference is  $O_k(M^2/n)$  uniformly for  $s \leq M$ .  $\square$

## 5 The exact scaling limit

**Theorem 5.1** (Existence and identification of the large-horizon limit). *For every fixed  $k \geq 1$ ,*

$$\boxed{\lim_{n \rightarrow \infty} \gamma_{n,k} = \rho_k}, \quad (27)$$

where  $\rho_k$  is the continuous value in (20).

*Proof.* We prove the two inequalities separately.

*Upper bound.* Choose a subsequence, still denoted  $n$ , along which  $\gamma_{n,k} \rightarrow \limsup_m \gamma_{m,k}$ , and choose policies  $(x^n, y^n)$  whose guarantees are at least  $\gamma_{n,k} - 1/n$ . By Lemma 4.2, the associated functions  $\tilde{C}_n \in \mathcal{C}_k$  satisfy  $\tilde{C}_n - C_{n;x^n,y^n} \rightarrow 0$  uniformly on every compact interval. Compactness of  $\mathcal{C}_k$  yields a further subsequence converging locally uniformly to some  $C \in \mathcal{C}_k$ .

Fix  $s > 0$ . For all sufficiently large  $n$ ,  $s \leq n$ , and therefore

$$C_{n;x^n,y^n}(s) \geq (\gamma_{n,k} - 1/n)B_{n,k}(s/n).$$

Using Lemmas 4.2 and 4.3 and passing to the limit gives

$$C(s) \geq \left( \limsup_m \gamma_{m,k} \right) Q_k(s).$$

Since this holds for every  $s > 0$ ,  $\rho_k \geq \limsup_n \gamma_{n,k}$ .

*Lower bound.* Fix  $\eta > 0$ , and let  $C^* \in \mathcal{C}_k$  attain (20), so

$$C^*(s) \geq \rho_k Q_k(s) \quad (s > 0). \quad (28)$$

Choose  $\varepsilon > 0$  and  $M < \infty$  such that

$$\frac{Q_k(\varepsilon)}{\varepsilon} \geq 1 - \eta, \quad \frac{Q_k(M)}{k} \geq 1 - \eta. \quad (29)$$

Choose a tolerance

$$0 < \tau \leq \eta \min\{Q_k(\varepsilon), \varepsilon, k\}. \quad (30)$$

By (18) and Lemma 3.3, there is a bounded step control  $a$  satisfying

$$C_a(s) \geq C^*(s) - \tau \quad (\varepsilon \leq s \leq M). \quad (31)$$



For  $n$  larger than the hazard bound, define  $p_i^\ell = a_\ell((i-1)/n)/n$ , and let  $(x^n, y^n)$  be the exact discrete occupancy flow with  $y_i^{n,\ell} = p_i^\ell x_i^{n,\ell}$ . Explicit Euler convergence for the linear state system, together with

$$C_{n;x^n,y^n}(s) = \frac{1}{n} \sum_{i=1}^n \sum_{\ell=1}^k x_i^{n,\ell} \min\{a_\ell((i-1)/n), s\}, \quad (32)$$

gives uniform convergence to  $C_a(s)$  on  $[0, M]$ . Lemma 4.3 gives uniform denominator convergence there. Since  $Q_k$  is bounded away from zero on  $[\varepsilon, M]$ , for all sufficiently large  $n$ ,

$$\inf_{\varepsilon \leq s \leq M} \frac{C_{n;x^n,y^n}(s)}{B_{n,k}(s/n)} \geq \rho_k - 3\eta. \quad (33)$$

Here we used (28), (31), and  $\tau/Q_k(\varepsilon) \leq \eta$ .

For  $0 < s \leq \varepsilon$ , Lemma 4.1, convergence at  $\varepsilon$ , and (29) give

$$\begin{aligned} \frac{C_{n;x^n,y^n}(s)}{B_{n,k}(s/n)} &\geq \frac{C_{n;x^n,y^n}(\varepsilon)}{\varepsilon} \\ &\geq \rho_k \frac{Q_k(\varepsilon)}{\varepsilon} - \frac{\tau}{\varepsilon} - o(1) \geq \rho_k - 2\eta - o(1), \end{aligned}$$

where  $\rho_k \leq 1$  was used. Likewise, for  $M \leq s \leq n$ ,

$$\begin{aligned} \frac{C_{n;x^n,y^n}(s)}{B_{n,k}(s/n)} &\geq \frac{C_{n;x^n,y^n}(M)}{k} \\ &\geq \rho_k \frac{Q_k(M)}{k} - \frac{\tau}{k} - o(1) \geq \rho_k - 2\eta - o(1). \end{aligned}$$

Combining the three ranges gives  $\liminf_n \gamma_{n,k} \geq \rho_k - 3\eta$ . Letting  $\eta \downarrow 0$  completes the proof.  $\square$

**Remark 5.2** (What closes the earlier scaling gap). *The proof uses only compact-scale convergence. The global conclusion follows from the two explicit anchor inequalities (21)–(22). No unsupported interchange of an infimum with pointwise limits is used.*

## 6 Compact-scale minimax duality and HJB

Fix  $0 < \varepsilon < M < \infty$  and define

$$\rho_k(\varepsilon, M) := \max_{C \in \mathcal{C}_k} \min_{s \in [\varepsilon, M]} \frac{C(s)}{Q_k(s)}. \quad (34)$$

For  $\nu \in \mathcal{P}([\varepsilon, M])$ , let

$$\Phi_\nu(a) := \int_{[\varepsilon, M]} \frac{\min\{a, s\}}{Q_k(s)} \nu(ds), \quad a \geq 0. \quad (35)$$

Because  $\Phi_\nu$  is constant for  $a \geq M$  and the value is nondecreasing in the inventory state, actions larger than  $M$  are never needed. Define  $u_0 \equiv 0$ ,  $u_\ell(1) = 0$ , and

$$-\dot{u}_\ell(t) = \max_{0 \leq a \leq M} \{ \Phi_\nu(a) - a(u_\ell(t) - u_{\ell-1}(t)) \}, \quad \ell = 1, \dots, k. \quad (36)$$

The right-hand side is globally Lipschitz in  $u$ , so this terminal-value system has a unique absolutely continuous solution.

**Theorem 6.1** (Compact-scale saddle formula). *For every  $0 < \varepsilon < M < \infty$ ,*

$$\boxed{\rho_k(\varepsilon, M) = \min_{\nu \in \mathcal{P}([\varepsilon, M])} u_k^\nu(0).} \quad (37)$$

*The minimum is attained.*

*Proof.* For  $C \in \mathcal{C}_k$  and  $\nu \in \mathcal{P}([\varepsilon, M])$ , set

$$L(C, \nu) := \int_{[\varepsilon, M]} \frac{C(s)}{Q_k(s)} \nu(ds).$$

This payoff is continuous and affine in each variable when  $\mathcal{C}_k$  has the compact-open topology and the measure space has the weak topology. Sion's theorem [7] gives

$$\rho_k(\varepsilon, M) = \min_{\nu \in \mathcal{P}([\varepsilon, M])} \max_{C \in \mathcal{C}_k} L(C, \nu). \quad (38)$$

The measure space is weakly compact, and the support function is lower semicontinuous, so the minimum is attained.

The inner support function is unchanged if one maximizes over ordinary bounded controls: taking a convex hull and then a closure does not change the supremum of a continuous linear functional. For a control  $a$  with state  $x$ , Fubini gives

$$L(C_a, \nu) = \sum_{\ell=1}^k \int_0^1 x_\ell(t) \Phi_\nu(a_\ell(t)) dt.$$

This is a finite-state continuous-time pure-death control problem. In state  $\ell$ , action  $a$  earns instantaneous reward  $\Phi_\nu(a)$  and jumps to  $\ell - 1$  at rate  $a$ . Dynamic programming gives (36), and its value from state  $k$  at time zero is  $u_k^\nu(0)$ . Substitution into (38) proves the theorem.  $\square$

## 7 An attained global compactified dual

Let

$$\overline{\mathbb{R}}_+ := [0, \infty]$$

have the compact topology induced by  $s \mapsto s/(1+s)$ . Let  $\mathcal{A}_+$  denote the class of bounded controls for which there is a constant  $c_a > 0$  satisfying  $a_\ell(t) \geq c_a$  for all  $\ell$  and almost every  $t$ .

**Lemma 7.1** (Strictly positive hazards preserve the primal value). *One has*

$$\rho_k = \sup_{a \in \mathcal{A}_+} \inf_{s > 0} \frac{C_a(s)}{Q_k(s)}. \quad (39)$$

*Proof.* The right-hand side is at most  $\rho_k$  because every  $C_a$  belongs to  $\mathcal{C}_k$ . For the reverse inequality, let  $C^*$  be a primal maximizer and fix  $\eta > 0$ . Choose  $\varepsilon, M$  as in (29). By (18), select a bounded control  $a$  whose coverage is within an arbitrarily small error from below on  $[\varepsilon, M]$ . Replace each hazard by  $a_\ell + c$ . Lemma 3.3 shows that coverage changes locally uniformly by  $o(1)$  as  $c \downarrow 0$ . For sufficiently small approximation and perturbation errors, the ratio on  $[\varepsilon, M]$  is at least  $\rho_k - O(\eta)$ . Lemma 4.1, applied at the anchors  $\varepsilon$  and  $M$ , gives the same global lower bound. Letting  $\eta \downarrow 0$  proves the result.  $\square$

For  $a \in \mathcal{A}_+$ , the ratio

$$R_a(s) := \frac{C_a(s)}{Q_k(s)}, \quad s > 0, \quad (40)$$

extends continuously to  $\overline{\mathbb{R}}_+$  by

$$R_a(0) = \int_0^1 \sum_{\ell=1}^k x_\ell(t) dt, \quad (41)$$

$$R_a(\infty) = \frac{1}{k} \sum_{\ell=1}^k \int_0^1 x_\ell(t) a_\ell(t) dt. \quad (42)$$

Indeed, for sufficiently small  $s$ ,  $\min\{a_\ell, s\} = s$ , while for sufficiently large  $s$ ,  $\min\{a_\ell, s\} = a_\ell$ ; then use (5). Lemma 3.4 implies

$$0 \leq R_a(s) \leq 1 \quad (s \in \overline{\mathbb{R}}_+). \quad (43)$$

For  $\nu \in \mathcal{P}(\overline{\mathbb{R}}_+)$ , define

$$J_k(\nu) := \sup_{a \in \mathcal{A}_+} \int_{\overline{\mathbb{R}}_+} R_a(s) \nu(ds). \quad (44)$$

**Lemma 7.2** (Terminal completion removes mass at infinity). *If  $\nu = \beta\delta_\infty + (1 - \beta)\bar{\nu}$  with  $0 \leq \beta < 1$  and  $\bar{\nu} \in \mathcal{P}([0, \infty))$ , then*

$$J_k(\nu) \geq \beta + (1 - \beta)J_k(\bar{\nu}) \geq J_k(\bar{\nu}). \quad (45)$$

*Proof.* Choose  $a \in \mathcal{A}_+$  nearly optimal for  $\bar{\nu}$ . For  $\delta \in (0, 1)$ , run its time compression on  $[0, 1 - \delta]$ :

$$a_\ell^\delta(t) := \frac{1}{1 - \delta} a_\ell\left(\frac{t}{1 - \delta}\right).$$

The state is  $x_\ell^\delta(t) = x_\ell(t/(1 - \delta))$ . The contribution of this first segment at a finite positive scale  $s$  is exactly  $C_a((1 - \delta)s)$ , divided by  $Q_k(s)$  after normalization. It converges pointwise to  $R_a(s)$  and is bounded by  $Q_k((1 - \delta)s)/Q_k(s) \leq 1$ . At scale zero it equals  $(1 - \delta)R_a(0)$ . Dominated convergence therefore shows that its  $\bar{\nu}$ -payoff converges to the original payoff.

On  $[1 - \delta, 1]$ , use the same constant hazard  $h_\delta$  in every positive state, with  $h_\delta \delta \rightarrow \infty$ . Conditional on having  $r$  slots left at time  $1 - \delta$ , the number of terminal transitions is distributed as  $\min\{\text{Pois}(h_\delta \delta), r\}$ , so all remaining selections are completed with probability tending to one. Hence the endpoint reward  $R(\infty)$  tends to one. All finite-scale terminal rewards are nonnegative. Taking limits and then optimizing over the initial control gives the first inequality. The second follows from  $J_k(\bar{\nu}) \leq 1$ .  $\square$

**Theorem 7.3** (Attained compactified dual). *There exists  $\nu^* \in \mathcal{P}(\overline{\mathbb{R}}_+)$  such that*

$$\rho_k = \min_{\nu \in \mathcal{P}(\overline{\mathbb{R}}_+)} J_k(\nu) = J_k(\nu^*). \quad (46)$$

*Moreover, an optimal measure can be chosen with no mass at  $\infty$ .*

*Proof.* Let  $K_j = [1/j, j]$  and  $\rho_j := \rho_k(1/j, j)$ . Then  $\rho_j$  decreases, and

$$\rho_j \downarrow \rho_k. \quad (47)$$

Indeed,  $\rho_j \geq \rho_k$ . Conversely, choose maximizers  $C_j \in \mathcal{C}_k$ . By compactness, pass to a locally uniformly convergent subsequence with limit  $C^* \in \mathcal{C}_k$ . For every fixed  $m$  and all  $j \geq m$ ,

$$\min_{s \in K_m} \frac{C_j(s)}{Q_k(s)} \geq \rho_j.$$

Passing to the limit and then letting  $m \rightarrow \infty$  gives  $\inf_{s>0} C^*(s)/Q_k(s) \geq \lim_j \rho_j$ , proving (47).

By Theorem 6.1, choose an optimal  $\nu_j \in \mathcal{P}(K_j)$  and regard it as a measure on  $\overline{\mathbb{R}}_+$ . Along a subsequence,  $\nu_j \Rightarrow \nu^*$ . For every fixed  $a \in \mathcal{A}_+$ , the extended  $R_a$  is continuous, and

$$\int R_a d\nu_j \leq \rho_j.$$

Therefore

$$\int R_a d\nu^* = \lim_j \int R_a d\nu_j \leq \rho_k.$$

Taking the supremum over  $a$  gives  $J_k(\nu^*) \leq \rho_k$ .

Conversely, for every  $\nu \in \mathcal{P}(\overline{\mathbb{R}}_+)$ ,

$$J_k(\nu) \geq \sup_{a \in \mathcal{A}_+} \min_{s \in \overline{\mathbb{R}}_+} R_a(s) = \rho_k.$$

The equality follows from Lemma 7.1; for a continuous extended ratio, the minimum on the compactification equals the infimum over positive finite scales. Thus  $J_k(\nu^*) = \rho_k$ .

Finally, if an optimizer has mass  $\beta \in (0, 1)$  at infinity, Lemma 7.2 shows that deleting and renormalizing that mass cannot increase  $J_k$ . If the optimizer is  $\delta_\infty$ , then  $J_k = 1$ . Either  $\rho_k < 1$  and it is not optimal, or  $\rho_k = 1$ , in which case any finite-scale Dirac measure is also optimal by Lemma 3.4. Hence an optimizer with no mass at infinity always exists.  $\square$

**Remark 7.4.** *The proof does not infer a saddle from compactness alone. It constructs  $\nu^*$  as a weak limit of genuine compact-scale minimizers and uses weak convergence only against fixed continuous ratio functions.*

## 8 The exact measure–HJB and tail formulas

Choose a dual measure with no mass at infinity and write

$$\nu = \kappa \delta_0 + \mu, \quad \kappa \geq 0, \quad \mu \in \mathcal{M}_+((0, \infty)), \quad \kappa + \mu((0, \infty)) = 1. \quad (48)$$

For  $a \geq 0$ , define

$$\Phi_\nu(a) := \kappa + \int_{(0, \infty)} \frac{\min\{a, s\}}{Q_k(s)} \mu(ds). \quad (49)$$

The atom at zero contributes live-state occupancy reward. For each finite  $A$ , let  $u^{A, \nu}$  be the unique solution of

$$-\dot{u}_\ell^{A, \nu}(t) = \max_{0 \leq a \leq A} \left\{ \Phi_\nu(a) - a(u_\ell^{A, \nu}(t) - u_{\ell-1}^{A, \nu}(t)) \right\}, \quad u_\ell^{A, \nu}(1) = 0, \quad u_0^{A, \nu} \equiv 0. \quad (50)$$

These values increase with  $A$  and are bounded by one. Indeed, at each positive scale the value from state  $\ell$  with remaining horizon  $\tau \leq 1$  is at most  $Q_\ell(s\tau)/Q_k(s) \leq 1$ , and the zero-scale occupancy reward is at most one. Define the value solution

$$u_\ell^\nu(t) := \lim_{A \rightarrow \infty} u_\ell^{A, \nu}(t). \quad (51)$$

No classical differentiability at a possible terminal singularity is asserted.

**Theorem 8.1** (Exact measure–HJB formula). *For every fixed  $k$ ,*

$$\rho_k = \min_{\substack{\kappa \geq 0, \mu \geq 0 \\ \kappa + \mu((0, \infty)) = 1}} u_k^{\kappa, \mu}(0). \quad (52)$$

*The minimum is attained.*

*Proof.* For  $a \in \mathcal{A}_+$ , Fubini and (41) give

$$\int_{\mathbb{R}_+} R_a(s) \nu(ds) = \sum_{\ell=1}^k \int_0^1 x_\ell(t) \Phi_\nu(a_\ell(t)) dt.$$

With actions restricted to  $[0, A]$ , standard finite-state dynamic programming gives (50). Allowing zero actions does not enlarge the closure incorrectly: replacing any nonnegative bounded control by  $a + c$  and sending  $c \downarrow 0$  gives convergence of states and positive-scale rewards, while the zero-scale reward converges to the constant  $\kappa$  in (49). Thus the union over  $A$  has value  $J_k(\nu)$ , and (51) computes it. Theorem 7.3 supplies an optimal measure with no mass at infinity, proving (52) and attainment.  $\square$

We now rewrite the positive-scale measure by a decreasing weighted tail.

**Definition 8.2** (Admissible tails). *Let  $\mathfrak{H}_k$  be the set of pairs  $(\kappa, H)$  such that*

- (i)  $\kappa \geq 0$ ;
- (ii)  $H : (0, \infty) \rightarrow [0, \infty)$  is nonincreasing and right-continuous, with  $H(r) \rightarrow 0$  as  $r \rightarrow \infty$ ;
- (iii)  $H$  is integrable on every interval  $(0, a)$ ; and
- (iv)

$$\kappa + \int_0^\infty w_k(r) H(r) dr = 1. \quad (53)$$

For  $(\kappa, H) \in \mathfrak{H}_k$ , put

$$\Phi_{\kappa, H}(a) := \kappa + \int_0^a H(r) dr. \quad (54)$$

**Theorem 8.3** (Attained tail–Hamiltonian characterization). *For every fixed  $k \geq 1$ ,*

$$\rho_k = \min_{(\kappa, H) \in \mathfrak{H}_k} u_k^{\kappa, H}(0), \quad (55)$$

where  $u$  is the increasing bounded-action value limit for

$$-\dot{u}_\ell(t) = \sup_{a \geq 0} \left\{ \kappa + \int_0^a H(r) dr - a(u_\ell(t) - u_{\ell-1}(t)) \right\}, \quad u_\ell(1) = 0, \quad u_0 \equiv 0. \quad (56)$$

*The minimum is attained. Equation (56) is interpreted through bounded-action truncation as in (50)–(51).*

*Proof.* Given (48), define

$$H(r) := \int_{(r,\infty)} \frac{1}{Q_k(s)} \mu(ds). \quad (57)$$

Tonelli gives, for every  $a > 0$ ,

$$\int_0^a H(r) dr = \int_{(0,\infty)} \frac{\min\{a, s\}}{Q_k(s)} \mu(ds), \quad (58)$$

$$\int_0^\infty w_k(r) H(r) dr = \int_{(0,\infty)} \frac{Q_k(s)}{Q_k(s)} \mu(ds) = \mu((0, \infty)). \quad (59)$$

Thus  $(\kappa, H) \in \mathfrak{H}_k$  and the Hamiltonians agree.

Conversely, let  $(\kappa, H) \in \mathfrak{H}_k$ . Define a Stieltjes measure on  $(0, \infty)$  by

$$\mu(ds) := -Q_k(s) dH(s). \quad (60)$$

Local integrability and monotonicity imply  $sH(s) \rightarrow 0$  as  $s \downarrow 0$ ; since  $Q_k(s) \sim s$ , the lower boundary term vanishes. Since  $H(s) \rightarrow 0$  and  $Q_k(s) \rightarrow k$ , the upper boundary term also vanishes. Stieltjes integration by parts therefore gives

$$\mu((0, \infty)) = \int_0^\infty w_k(s) H(s) ds = 1 - \kappa.$$

A second integration by parts yields

$$\int_{(0,\infty)} \frac{\min\{a, s\}}{Q_k(s)} \mu(ds) = \int_{(0,\infty)} \min\{a, s\} (-dH(s)) = \int_0^a H(r) dr.$$

Thus the measure and tail formulations are equivalent. Attainment follows from Theorem 8.1.  $\square$

## 9 Conditional complementary slackness and a regular system

The measure–HJB formula is unconditional. A classical Hill–Kertz-type differential system requires additional properties that are not proved here for  $k \geq 2$ .

**Assumption 9.1** (Regular saddle branch). *Assume that an optimal pair  $(\kappa, H)$  and a primal continuous-time control satisfy all of the following:*

- (i) *the control attains the primal value  $\rho_k$  and maximizes the weighted objective of the optimal dual pair;*
- (ii)  *$H \in C^1(0, \infty)$ ,  $H'(s) < 0$ , and the maximizing actions are finite on  $[0, 1)$ ;*
- (iii) *the positive-scale dual measure has full support on  $(0, \infty)$ ; and*
- (iv) *the state and action functions have enough regularity for the differentiations below.*

Under this assumption, put

$$\mathcal{R}(a) := \kappa + \int_0^a H(r) dr - aH(a). \quad (61)$$

The first-order condition in the Hamiltonian is

$$H(a_\ell(t)) = u_\ell(t) - u_{\ell-1}(t). \quad (62)$$

Differentiating and using the HJB equations gives

$$H'(a_\ell(t))\dot{a}_\ell(t) = \mathcal{R}(a_{\ell-1}(t)) - \mathcal{R}(a_\ell(t)), \quad \ell = 1, \dots, k, \quad (63)$$

with the convention  $\mathcal{R}(a_0) = 0$ . The forward state equation is (13).

Complementary slackness says that the coverage ratio equals  $\rho_k$  on the support of the optimal measure. Full support therefore yields

$$\sum_{\ell=1}^k \int_0^1 x_\ell(t) \min\{a_\ell(t), s\} dt = \rho_k Q_k(s), \quad s > 0. \quad (64)$$

Differentiating at regular values of  $s$  gives

$$\boxed{\sum_{\ell=1}^k \int_{\{t: a_\ell(t) > s\}} x_\ell(t) dt = \rho_k w_k(s), \quad s > 0.} \quad (65)$$

**Proposition 9.2** (Conditional regular equations). *Under Assumption 9.1, every regular saddle satisfies (13), (63), (65), and the normalization (53). Conversely, a sufficiently regular feasible control and tail pair satisfying HJB stationarity, the global coverage inequalities, and complementary slackness certify the value  $\rho_k$  by weak duality.*

**Remark 9.3.** *Neither full support nor smoothness follows from Theorem 8.3. Therefore (63)–(65) are a conditional reduction, not an unconditional finite ODE characterization for  $k \geq 2$ .*

## 9.1 The case $k = 1$

For  $k = 1$ , the classical regular branch is known [3, 5]. Write  $a = a_1$ ,  $x = x_1$ . Then  $\dot{x} = -ax$ ,  $x(0) = 1$ , and  $Q_1(s) = 1 - e^{-s}$ ,  $w_1(s) = e^{-s}$ . The marginal equalizer becomes

$$\int_{\{t: a(t) > s\}} x(t) dt = \rho_1 e^{-s}. \quad (66)$$

For an increasing threshold, setting  $s = a(t)$  gives

$$\int_t^1 x(v) dv = \rho_1 e^{-a(t)}. \quad (67)$$

Let  $y(t) = e^{-a(t)}$ . Differentiation yields  $x = -\rho_1 y'$ , while the state equation gives  $x' = x \log y$ . Hence  $y'' = y' \log y$ . Using  $x(0) = 1$ ,  $a(0) = 0$ , and terminal blow-up gives

$$\boxed{y' = y \log y - y + 1 - \frac{1}{\rho_1}, \quad y(0) = 1, \quad y(1) = 0.} \quad (68)$$

This is the Hill–Kertz equation. Its unique admissible constant is  $\rho_1 \approx 0.7454403321$ , providing a numerical consistency check on all signs and normalizations.

## 10 What is and is not solved

Brustle–Perez-Salazar–Verdugo prove that their nonlinear splitting system yields asymptotic lower bounds and explicitly leave its tightness open [2]. They also leave open an analytic formula for the exact multi-selection ratio and whether

$$\inf_{n \geq k} \gamma_{n,k} = \sum_{\ell=1}^k \theta_{\ell}^*, \quad (69)$$

where the right-hand side is generated by their system.

The results proved here establish:

1. the limit  $\lim_{n \rightarrow \infty} \gamma_{n,k}$  exists for every fixed  $k$ ;
2. this limit has the exact attained measure–HJB and tail–Hamiltonian characterizations in Theorems 8.1 and 8.3;
3. every finite-horizon value  $\gamma_{n,k}$  is algebraic.

They do not establish:

1. that the BPSV splitting value equals  $\rho_k$ ;
2. that  $\inf_n \gamma_{n,k} = \rho_k$ ;
3. that an optimal dual measure is smooth or has full support;
4. that the conditional free-boundary system has a unique solution for  $k \geq 2$ .

Accordingly, this paper gives substantial partial progress, not a complete solution of the authors’ open problem.

## A Euler approximation used in Theorem 5.1

Let  $a$  be a bounded step control, with bound  $A$ , and let  $x$  solve (13). Define  $p_i^{\ell} = a_{\ell}((i-1)/n)/n$  for  $n > A$ , and let  $x_i^{n,\ell}$  satisfy the exact discrete flow. The state equation is a linear system  $\dot{x} = G(t)x$  with a bounded piecewise constant generator. Explicit Euler gives

$$\max_{i \leq n+1} \|x_i^n - x((i-1)/n)\|_1 = O_{A,k}(1/n). \quad (70)$$

For  $s \in [0, M]$ , equation (32) holds. The summand is uniformly bounded and is 1-Lipschitz in  $s$ . Equation (70) and ordinary Riemann-sum convergence therefore imply uniform convergence on  $[0, M]$ .

## B A compactness warning

The failed compactness inference in the earlier draft can be seen in the following elementary example. Let  $S = \mathbb{N} \cup \{\infty\}$  and define

$$R(j) = \frac{1}{j}, \quad R(\infty) = 1.$$



The singleton strategy set  $\{R\}$  is compact, but  $\inf_{s \in S} R(s) = 0$  is not attained. For every countably additive probability measure  $\nu$  on  $S$ ,  $\int R d\nu > 0$ , although the infimum over such measures is zero, approached by  $\delta_j$ . Thus compactness of the strategy spaces alone does not imply dual attainment. The proof of Theorem 7.3 avoids this pathology by deriving compact-scale minimizers first and passing to weak limits only against continuous endpoint-extended payoffs.

Likewise, product-topology convergence does not preserve a max–min guarantee. If

$$R_m(j) = \begin{cases} 0, & j = m, \\ 1, & j \neq m, \end{cases}$$

then  $R_m \rightarrow 1$  pointwise while  $\inf_j R_m(j) = 0$  for every  $m$ . The tail inequalities in Lemma 4.1 are the quantitative substitute needed in the present problem.

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